

Lesson 04: Introduction to Initial Value Problems (Particular Solutions & Curve Families)

Professor Leonard Differential Equations Master Series · Lesson 4

- **Direct Video Link:** [Lesson 04: Introduction to Initial Value Problems \(Particular Solutions & Curve Families\)](#) · Duration: 28:57
- **Target Exam Scope:** 🎯 MIDTERM EXAM (Ch 1.2 — Initial Value Problems)
- **Course Context:** Applied Ordinary Differential Equations (ENGR 213)

1. Whiteboard Lecture Intuition & Philosophical Framing

In this 29-minute lecture, Professor Leonard explores Initial Value Problems (IVPs). He demystifies the integration constant $+C$, proving that $+C$ represents geometric degrees of freedom. Providing an initial condition (x_0, y_0) acts as a spatial anchor, locking onto the single specific curve that passes through that point.

2. Core Theoretical Concepts & Mathematical Formulations

Anatomy of an Initial Value Problem (IVP)

An n -th order IVP consists of an ordinary differential equation paired with n initial conditions specified at **the exact same independent variable point** x_0 :

$$\begin{cases} a_n(x)y^{(n)} + \cdots + a_0(x)y = g(x) & y(x_0) = y_0, & y'(x_0) = y_1, & \dots, & y^{(n-1)}(x_0) = y_{n-1} \end{cases}$$

Why Are n Conditions Required for an n -th Order ODE?

- Every integration step introduces one independent constant of integration:
 - 1st-order ODE $\implies$ 1 integration $\implies$ 1 constant $c_1 \implies$ 1 condition $y(x_0) = y_0$.
 - 2nd-order ODE $\implies$ 2 integrations $\implies$ 2 constants $c_1, c_2 \implies$ 2 conditions $y(x_0) = y_0, y'(x_0) = y_1$.
 - An IVP differs from a **Boundary Value Problem (BVP)** where conditions are specified at different points (e.g., $y(0) = 0$ and $y(L) = 0$).
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3. Classroom Chalkboard Examples (Solved in Complete Baby Steps)

Problem 04.1: Chalkboard Problem 4.1: First-Order Linear IVP with Initial Anchor

Problem Statement:

Given that the one-parameter family $y(x) = Ce^{-3x} + \frac{1}{3}x - \frac{1}{9}$ is the general solution to $y' + 3y = x$, find the particular solution satisfying the initial condition $y(0) = 2$.

Step-by-Step Whiteboard Solution:

- **Step 1: Substitute the Initial Condition ($x = 0, y = 2$) into the General Solution:**

$$y(0) = Ce^{-3(0)} + \frac{1}{3}(0) - \frac{1}{9} = 2$$

$$C(1) + 0 - \frac{1}{9} = 2 \implies C - \frac{1}{9} = 2$$

- **Step 2: Solve Algebraically for Constant C :**

$$C = 2 + \frac{1}{9} = \frac{18}{9} + \frac{1}{9} = \frac{19}{9}$$

- **Step 3: Write Down the Particular Solution:**

$$y_p(x) = \frac{19}{9}e^{-3x} + \frac{1}{3}x - \frac{1}{9}$$

- **Step 4: Verify the Particular Solution and Initial Condition:**

- Check initial condition: $y_p(0) = \frac{19}{9}(1) + 0 - \frac{1}{9} = \frac{18}{9} = 2 \quad \checkmark$
- Check differential equation:

$$y'_p(x) = -3 \left(\frac{19}{9} \right) e^{-3x} + \frac{1}{3} = -\frac{19}{3}e^{-3x} + \frac{1}{3}$$

$$y'_p + 3y_p = \left(-\frac{19}{3}e^{-3x} + \frac{1}{3} \right) + 3 \left(\frac{19}{9}e^{-3x} + \frac{1}{3}x - \frac{1}{9} \right)$$

$$y'_p + 3y_p = -\frac{19}{3}e^{-3x} + \frac{1}{3} + \frac{19}{3}e^{-3x} + x - \frac{1}{3} = x = \text{RHS} \quad \checkmark$$

[!WARNING] **Common Exam Pitfall:** Forgetting to evaluate $e^0 = 1$. Students often accidentally set $e^0 = 0$, leading to completely invalid calculations for C .

Problem 04.2: Chalkboard Problem 4.2: Second-Order Kinematic IVP with Position and Velocity Anchors

Problem Statement:

Solve the 2nd-order IVP: $y'' = 12x^2 - 4$, subject to $y(1) = 3$ and $y'(1) = 8$.

Step-by-Step Whiteboard Solution:

- **Step 1: First Integration to Obtain Velocity $y'(x)$:**

$$y'(x) = \int (12x^2 - 4)dx = 4x^3 - 4x + C_1$$

- **Step 2: Apply First Initial Condition $y'(1) = 8$ to Solve for C_1 Immediately:**

$$y'(1) = 4(1)^3 - 4(1) + C_1 = 8 \implies 4 - 4 + C_1 = 8 \implies C_1 = 8$$

Thus: $y'(x) = 4x^3 - 4x + 8$.

- **Step 3: Second Integration to Obtain Position $y(x)$:**

$$y(x) = \int (4x^3 - 4x + 8)dx = x^4 - 2x^2 + 8x + C_2$$

- **Step 4: Apply Second Initial Condition $y(1) = 3$ to Solve for C_2 :**

$$y(1) = (1)^4 - 2(1)^2 + 8(1) + C_2 = 3$$

$$1 - 2 + 8 + C_2 = 3 \implies 7 + C_2 = 3 \implies C_2 = -4$$

- **Step 5: Assemble the Unique Particular Solution:**

$$y_p(x) = x^4 - 2x^2 + 8x - 4$$

- **Step 6: Comprehensive Verification:**

- $y_p(1) = 1 - 2 + 8 - 4 = 3 \quad \checkmark$
- $y'_p(x) = 4x^3 - 4x + 8 \implies y'_p(1) = 4 - 4 + 8 = 8 \quad \checkmark$
- $y''_p(x) = 12x^2 - 4 \quad \checkmark$

[!WARNING] **Common Exam Pitfall:** Attempting to integrate twice before solving for C_1 , which produces $C_1x + C_2$ and increases algebraic complexity. Solve for constants immediately as each derivative condition is satisfied!

4. Key Takeaways & Exam Strategy Formula Card

- Watch the full whiteboard demonstration on YouTube: [Lesson 04: Introduction to Initial Value Problems \(Particular Solutions & Curve Families\)](#).
- Master the classification and verification mechanics before attempting complex integrations.
- Always check domain restrictions and verify initial values back into the computed trajectory.